

SAMS Operations Runbook

Smart Attendance Management System — Super Admin troubleshooting blueprint

Item	Value
Production app	https://app.smart-managment.com
Super Admin	https://super.smart-managment.com (school code SUPERADMIN)
API (direct)	https://api.smart-managment.com
VPS path	/var/www/sams
API port (local)	3001
PM2 app name	sams-api

Use this when something breaks: **symptom** → **where to check** → **commands** → **fix**.

1. First response (any outage)

```
cd /var/www/sams
pm2 status
curl -sS http://127.0.0.1:3001/health
sudo nginx -t
```

Health field	Meaning
checks.database: true	PostgreSQL OK
checks.redis: true	Redis OK
ai.configured: true	AI keys loaded
sms.configured: true	Africa's Talking keys loaded

If `/health` fails locally → **backend** issue (PM2, env, DB, Redis).

If local OK but browser fails → **nginx** or **Cloudflare/DNS**.

2. Site down / blank page / 502

Symptom: White screen or 502 Bad Gateway

Check: PM2 running?

```
pm2 status
pm2 logs sams-api --lines 50 --nostream
```

Fix — start API:

```
cd /var/www/sams
pm2 start ecosystem.config.js --env production --update-env
pm2 save
bash scripts/restart-api.sh
```

Check: nginx:

```
sudo nginx -t
sudo systemctl status nginx
sudo systemctl reload nginx
```

Check: Frontend built?

```
ls packages/frontend/dist/index.html
ls packages/super-admin/dist/index.html
```

Fix — rebuild UI:

```
cd /var/www/sams
npm run build -w @sams/frontend
npm run build -w @sams/super-admin
```

3. PM2 empty after deploy

Symptom: `pm2 status` shows no `sams-api` after `deploy-production.sh`

Cause: Often junk line in `secrets/providers.env` (e.g. pasted `bash scripts/restart-api.sh#`) executed during env load.

Check:

```
grep -n "restart-api|bash \|pm2 \|curl " secrets/providers.env packages/backend/.env
```

Fix env file: Only `#` comments and `KEY="value"` lines. Then:

```
bash scripts/restart-api.sh
```

Never paste: `bash scripts/restart-api.sh#` (trailing `#` breaks the path).

4. JWT / API crash loop

Symptom: Logs show `JWT_SECRET` must be ... 64+ chars

```
bash scripts/set-production-env.sh
bash scripts/restart-api.sh
```

Check:

```
bash scripts/verify-secrets.sh
```

5. Redis errors

Symptom: `%22redis://localhost:6379%22` or `Connection is closed`

Cause: `REDIS_URL` has extra quote characters in env file.

Check:

```
grep REDIS_URL packages/backend/.env secrets/providers.env
redis-cli ping
sudo systemctl status redis-server
```

Fix — correct line (no doubled quotes):

```
REDIS_URL=redis://localhost:6379
```

```
sudo systemctl start redis-server
bash scripts/restart-api.sh
```

Good log line: `[Redis] Connected` and `Database and Redis connected – API ready` .

6. Database / migrations

Symptom: `PlanTier already exists / migration P3018`

Cause: Baseline migration on existing DB.

```
cd /var/www/sams/packages/backend
npx prisma migrate resolve --rolled-back 20240101000000_init
npx prisma migrate resolve --applied 20240101000000_init
npx prisma migrate deploy
```

Or use:

```
cd /var/www/sams
bash scripts/baseline-prisma-init.sh
cd packages/backend && npx prisma migrate deploy
```

Symptom: `Can't connect to database`

```
grep DATABASE_URL packages/backend/.env
sudo systemctl status postgresql
```

7. Nginx / uploads / 413 photo error

Symptom: AI says "Could not send that photo" / HTTP 413

Cause: nginx `client_max_body_size` too small on **app** or **super-admin** (not only api subdomain).

Check:

```
grep -n client_max_body_size /etc/nginx/sites-enabled/sams.conf
```

Need **3 lines** at `25m` (app, api, super).

Fix:

```
cd /var/www/sams
git pull origin main
sudo cp nginx/sams.conf /etc/nginx/sites-enabled/sams.conf
sudo nginx -t && sudo systemctl reload nginx
npm run build -w @sams/frontend
npm run build -w @sams/super-admin
```

8. AI not working

Symptom: Generic AI errors / not configured

```
bash scripts/verify-secrets.sh --ai-only
bash scripts/diagnose-ai.sh
curl -sS http://127.0.0.1:3001/health | grep -o '"ai":{[^}]*}'
```

Keys live in: `secrets/providers.env` (not git)

Required: `OPENAI_API_KEY` , `OPENAI_BASE_URL` , `OPENAI_MODEL` , optional fallback, `VISION_MODEL` for images.

Groq-only: Image upload needs OpenRouter fallback — Groq text models don't read images.

```
bash scripts/restart-api.sh
```

Symptom: AI talks but won't execute actions (suspend/unsuspend)

- Use explicit commands: `unsuspend school [name]` , `suspend school [name]`
- Destructive actions need reply **yes**
- Super Admin only for platform actions

Symptom: Super Admin AI vs school AI

Panel	URL	Scope
Super Admin AI	super.smart-managment.com	All schools, licenses, suspend
School AI	app.smart-managment.com	One school only

9. SMS / OTP

Symptom: SMS not delivered

```
curl -sS http://127.0.0.1:3001/health | grep -o '"sms":{[^}]*}'
```

Check providers.env :

- AT_USERNAME=deno1 (dashboard app username, **not** sams)
- AT_API_KEY=atsk_...
- AT_SENDER_ID=SAMS (must be approved in Africa's Talking dashboard)

OTP flags:

- OTP_LOGIN_ENABLED=false (login without SMS)
- OTP_PASSWORD_RESET_ENABLED=true (forgot-password SMS when sender approved)

Until AT approves sender **SAMS**, SMS will fail with InvalidSenderId — config can still be correct.

```
bash scripts/configure-production-at.sh
bash scripts/restart-api.sh
```

10. Email

Symptom: [Email] SMTP not configured

Optional. Set in providers.env : SMTP_HOST , SMTP_USER , SMTP_PASS , etc.

In-app notifications work without email.

11. School suspended / users can't login

Symptom: SCHOOL_SUSPENDED / 403

Super Admin — unsuspend via AI:

```
unsuspend school [School Name]
```

Or Super Admin panel → school → Unsuspend.

Check DB:

```
cd /var/www/sams/packages/backend
npx prisma studio
# or SQL: SELECT name, "isSuspended" FROM "School";
```

12. Secrets & env files

File	Purpose	In git?
packages/backend/.env	JWT, DATABASE_URL, PORT	No
secrets/providers.env	AI, AT, SMTP, M-Pesa keys	No
secrets/providers.env.example	Template	Yes

Load order: `.env` → `.env.secrets` → `providers.env` (later wins).

Before editing secrets:

```
bash scripts/backup-secrets.sh
# or full backup:
bash scripts/backup-production.sh --with-uploads
```

Repair corrupted providers file:

```
bash scripts/fix-providers-env.sh
bash scripts/verify-secrets.sh
bash scripts/restart-api.sh
```

Rules:

- Never commit real keys
- Never paste shell commands into `.env` files
- Only `# comments` and `KEY="value"` lines

13. Deploy (full)

Before deploy:

```
cd /var/www/sams
bash scripts/backup-production.sh
```

Deploy:

```
bash scripts/deploy-production.sh
```

Deploy: backs up env → `git reset --hard origin/main` → rebuilds all packages → prisma migrate → PM2 start → nginx reload → post-deploy verify.

If deploy fails at restart:

```
pm2 start ecosystem.config.js --env production --update-env
pm2 save
```

Never use: `bash scripts/restart-api.sh#`

14. Backup & restore

Create backup:

```
bash scripts/backup-production.sh --with-uploads
ls -la backups/production-*
```

Download to PC (from Windows PowerShell):

```
scp -r root@182.143.228.182:/var/www/sams/backups/production-YYYYMMDD-HHMMSS
"$env:USERPROFILE\Desktop\SAMS-backups\"
```

Restore: See `RESTORE.md` inside backup folder.

15. Super Admin account

```
cd /var/www/sams/packages/backend
npm run create-super-admin
# Reset password:
SUPER_ADMIN_FORCE_RESET=true npm run create-super-admin
```

Login: <https://super.smart-managment.com> — code `SUPERADMIN` .

16. Log locations

Log	Path
PM2 stdout	<code>/var/log/sams/sams-api-out.log</code>
PM2 stderr	<code>/var/log/sams/sams-api-error.log</code>
PM2 live	<code>pm2 logs sams-api</code>
Nginx error	<code>/var/log/nginx/error.log</code>

17. Quick command cheat sheet

```
# Status
pm2 status
curl -sS http://127.0.0.1:3001/health

# Restart API (safe)
cd /var/www/sams && bash scripts/restart-api.sh

# Verify secrets
bash scripts/verify-secrets.sh
```

```
# Nginx
sudo nginx -t && sudo systemctl reload nginx

# Full deploy
bash scripts/backup-production.sh && bash scripts/deploy-production.sh

# Git update only
cd /var/www/sams && git pull origin main
```

18. Load this runbook into Super Admin AI

Option A — Automatic (after deploy): This file is injected into Super Admin AI context from `docs/SAMS-OPS-RUNBOOK.md`.

Option B — Knowledge Base (editable in UI):

1. Super Admin → AI Knowledge (or API `POST /api/v1/super/ai-knowledge`)
2. Category: `ops`
3. Title: `SAMS Operations Runbook`
4. Paste sections you use most (keep under ~8000 chars per entry if splitting)

Option C — PDF for your desk: See `docs/README-RUNBOOK-PDF.md`

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Last updated for VPS layout `/var/www/sams` , PM2 `sams-api` , nginx multi-subdomain setup.